

E-Commerce Sales Data Analysis Using Power Bi

Objective

The objective of this project was to analyze raw e-commerce transaction data to extract meaningful insights related to sales performance, customer behavior, and operational efficiency.

Dataset Overview

The dataset consisted of transactional e-commerce data with the following fields

- Customer Id
- Order Id
- Order Date
- Product
- Category
- Quantity
- Price
- Payment Method
- Status
- Total

The dataset contained inconsistencies, missing values and formatting issues, making it suitable for real-world data cleaning and analysis.

ID	Customer Name	Order ID	Order Date	Product	Category	Quantity	Price	Payment Method	Status	Total
100	Customer_100	ORD-41285	11/22/2024	Blender	Home	3	38	Cash on Delivery	Shipped	114
101	Customer_101	ORD-35783	7/5/2025	Smartphon	Electronics	2	abd	PayPal	Processing	
102	Customer_102	ORD-84355	12/3/2024	Tennis Rac	Sports	1	389.05	PayPal	Delivered	389.05
103	Customer_103	ORD-57811	3/19/2025	Science	Books	5	233.92	PayPal	Processing	1169.6
104	Customer_104	ORD-93614	10/20/2025	Biography	Books	1	552.51	Cash on Delivery	Processing	552.51
105	Customer_105	ORD-22442	11/20/2024	Tennis Rac	Sports	3	122.06	Cash on Delivery	Cancelled	366.18
106	Customer_106	ORD-25885	2/3/2025	Blender	Home		978.63	Bank Transfer	Processing	
107	Customer_107	ORD-89122	1/3/2025	Biography	Books	5	587.68	PayPal	Returned	2938.4
108	Customer_108	ORD-64400	10/23/2025	Science	Books	1	600.29	Cash on Delivery	Processing	600.29
109	Customer_109	ORD-18512	5/3/2025	Tennis Rac	Sports	5	188.34	Credit Card	Shipped	841.7
110	Customer_110	ORD-61020	9/26/2025	Basketball	sports	5	four hund	Cash on Delivery	Cancelled	
111	Customer_111	ORD-72349	10/22/2025	Comics	Books	2	865.77	Credit Card	Returned	1731.54
112	Customer_112	ORD-80867	3/2/2025	Shoes	Clothing	4	752.13	Bank Transfer	Delivered	3008.52
113	Customer_113	ORD-23994	7/24/2025	Fiction	Books	1	411.81	Credit Card	Processing	411.81
114	Customer_114	ORD-77417	Jan 5 2023	Smartphon	Electronics	2	413.13	Cash on Delivery	Shipped	826.26
115	Customer_115	ORD-72778	1/10/2025	Fiction	Books	2	796.84	Cash on Delivery	Returned	1115.576
116	Customer_116	ORD-63660	10/10/2025	Microwave	Home	4		Cash on Delivery	Processing	
117	Customer_117	ORD-72751	2/12/2025	Blender	electronic	-2	10000	Cash on Delivery	Processing	-20000
118	Customer_118	ORD-96061	5/31/2025	Science	Books	5	247.82	PayPal	Processing	1239.1
119	Customer_119	ORD-76016	12/19/2024	Basketball	Sports	4	438.19	Cash on Delivery	Processing	1752.76
120	Customer_120	ORD-14659	6/13/2025	Yoga Mat	ELECTRONIC	5	3005	Credit Card	Cancelled	
121	Customer_121	ORD-80067	10/11/2025	Basketball	Sports	4	523.01	Credit Card	Delivered	2092.04
122	Customer_122	ORD-21443	5/16/2025	Microwave	Home	2	556.93	PayPal	Shipped	1113.86
123	Customer_123	ORD-11314	10/6/2025	Vacuum	electronic	3	615.91	Cash on Delivery	Cancelled	1847.73
124	Customer_124	ORD-46136	5/31/2025	Comics	Books	5		PayPal	Cancelled	
125	Customer_125	ORD-23202	2/9/2025	Comics	Books	4	765.69	PayPal	Shipped	3062.76
126	Customer_126	ORD-11261	5/18/2025	Shoes	Clothing	4	836.8	Bank Transfer	Cancelled	3347.2
127	Customer_127	ORD-65482	5/12/2025	Blender	Home	5	336.02	Bank Transfer	Returned	1680.1
128	Customer_128	ORD-12270	11/10/2024	Lamp	electronics	3	898.69	Bank Transfer	Delivered	2696.07
129	Customer_129	ORD-48863	6/22/2025	Science	Books	4	736.71	Bank Transfer	Returned	2946.84
130	Customer_130	ORD-34007	8/15/2025	Tennis Rac	Sports	2		Bank Transfer	Processing	
131	Customer_131	ORD-91196	6/12/2025	Lamp	Home	5	616.63	PayPal	Shipped	3083.15
132	Customer_132	ORD-40186	4/2/2025	Yoga Mat	ELECTRONIC	1	266.7	PayPal	Delivered	266.7
133	Customer_133	ORD-68182	12/5/2024	Biography	nan	5	343.24	Credit Card	Shipped	1716.2
134	Customer_134	ORD-16585	10/4/2025	T-shirt	Clothing	-5	591.53	PayPal	Shipped	-2957.65
135	Customer_135	ORD-13656	2/9/2025	Smartwate	electronics	4	446.19	Cash on Delivery	Cancelled	1784.76
136	Customer_136	ORD-20985	6/12/2025	Headphones		1	696.71	Credit Card	Delivered	696.71
137	Customer_137	ORD-91254	2/23/2025	Vacuum	Home	3	-100	Cash on Delivery	Delivered	-300

Data Cleaning (Power Query Editor)

The raw dataset was cleaned and transformed using Power Query Editor.

- Converted data types (dates, numeric fields)
- Standardized categorical values (Category, Payment Method)
- Replaced inconsistent entries using “Replace Values”
- Handling missing values using fill down and fill up techniques
- Removed or filtered invalid/irrelevant records

The transformation steps were recorded in Power Query , ensuring a repeatable and automated cleaning process.

Table: TransformColumnTypes(*Replaced Value14*,CurrencyType)

V# ID	Customer Name	Order ID	Order Date	Product	Category	Quantity	Price	Payment Method
1	100 Customer_100	ORD-41295	11/22/2024	Blender	Home	3	38.00	Cash on Delivery
2	101 Customer_101	ORD-51783	7/9/2025	Smartphone	ELECTRONICS	2	0.00	PayPal
3	102 Customer_102	ORD-84355	12/23/2024	Tennis Racket	SPORTS	1	389.05	PayPal
4	103 Customer_103	ORD-57811	3/29/2025	Science	Books	5	233.92	PayPal
5	104 Customer_104	ORD-93614	10/20/2025	Biography	Books	1	552.51	Cash on Delivery
6	105 Customer_105	ORD-22442	11/20/2024	Tennis Racket	SPORTS	3	122.06	Cash on Delivery
7	106 Customer_106	ORD-25885	2/2/2025	Blender	Home	0	978.63	Bank Transfer
8	107 Customer_107	ORD-49122	1/20/2025	Biography	Books	3	567.68	PayPal
9	108 Customer_108	ORD-64400	10/29/2025	Science	Books	1	600.29	Cash on Delivery
10	109 Customer_109	ORD-18512	5/7/2025	Tennis Racket	SPORTS	5	168.34	Credit Card
11	110 Customer_110	ORD-61020	8/26/2025	Basketball	SPORTS	5	400.00	Cash on Delivery
12	111 Customer_111	ORD-72349	10/22/2025	Comics	Books	2	865.77	Credit Card
13	112 Customer_112	ORD-40987	8/2/2025	Shoes	Clothing	4	752.13	Bank Transfer
14	113 Customer_113	ORD-23994	7/24/2025	Fiction	Books	3	411.81	Credit Card
15	114 Customer_114	ORD-71417	1/2/2025	Smartphone	ELECTRONICS	2	412.13	Cash on Delivery
16	115 Customer_115	ORD-72778	1/20/2025	Fiction	Books	2	796.84	Cash on Delivery
17	116 Customer_116	ORD-63660	10/30/2025	Microwave	Home	4	0.00	Cash on Delivery
18	117 Customer_117	ORD-72751	2/12/2025	Blender	ELECTRONICS	-2	20,000.00	Cash on Delivery
19	118 Customer_118	ORD-96061	5/11/2025	Science	Books	5	247.82	PayPal
20	119 Customer_119	ORD-76016	12/19/2024	Basketball	SPORTS	4	438.19	Cash on Delivery
21	120 Customer_120	ORD-14659	4/21/2025	Yoga Mat	ELECTRONICS	5	300.00	Credit Card
22	121 Customer_121	ORD-40967	10/21/2025	Basketball	SPORTS	4	521.02	Credit Card
23	122 Customer_122	ORD-21443	5/16/2025	Microwave	Home	2	556.93	PayPal
24	123 Customer_123	ORD-11314	10/6/2025	Vacuum	ELECTRONICS	3	625.91	Cash on Delivery
25	124 Customer_124	ORD-46136	5/11/2025	Comics	Books	5	0.00	PayPal
26	125 Customer_125	ORD-23202	2/9/2025	Comics	Books	4	765.69	PayPal
27	126 Customer_126	ORD-11261	5/28/2025	Shoes	Clothing	4	836.80	Bank Transfer
28	127 Customer_127	ORD-45482	5/12/2025	Blender	Home	3	336.02	Bank Transfer
29	128 Customer_128	ORD-12270	11/20/2024	Lamp	ELECTRONICS	3	898.69	Bank Transfer
30	129 Customer_129	ORD-49863	6/22/2025	Science	Books	4	736.71	Bank Transfer
31	130 Customer_130	ORD-94007	8/15/2025	Tennis Racket	SPORTS	2	0.00	Bank Transfer
32	131 Customer_131	ORD-91396	6/11/2025	Lamp	Home	5	616.63	PayPal
33	132 Customer_132	ORD-40186	4/2/2025	Yoga Mat	ELECTRONICS	1	266.70	PayPal
34	133 Customer_133	ORD-68182	12/5/2024	Biography	ELECTRONICS	5	343.24	Credit Card
35	134 Customer_134	ORD-16585	10/4/2025	T-shirt	Clothing	-5	591.33	PayPal
36	135 Customer_135	ORD-13656	2/8/2025	Smartwatch	ELECTRONICS	4	446.19	Cash on Delivery
37	136 Customer_136	ORD-20985	6/12/2025	Headphones	ELECTRONICS	3	696.72	Credit Card
38	137 Customer_137	ORD-91254	2/21/2025	Vacuum	Home	3	-100.00	Cash on Delivery
39	138 Customer_138	ORD-14319	8/9/2025	Lamp	Home	1	531.06	PayPal
40	139 Customer_139	ORD-18418	2/9/2025	Smartphone	ELECTRONICS	2	681.38	Credit Card
41	140 Customer_140	ORD-67096	2/26/2025	Fiction	Books	4	588.98	Credit Card
42	141 Customer_141	ORD-45059	2/13/2025	Fiction	Books	3	546.04	Cash on Delivery
43	142 Customer_142	ORD-69018	10/30/2025	Shoes	Clothing	5	645.16	Credit Card

11 COLUMNS, 103 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 7:54 PM

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	ID	Customer_Name	Order_ID	Order_Date	Product	Category	Quantity	Price	Payment_Method	Status	Total															
2	100	Customer_100	ORD-41285	11/22/2024	Blender	Home	5	38	Cash on Delivery	Shipped	114															
3	101	Customer_101	ORD-35783	7/5/2025	Smartphone	ELECTRONICS	2	0	PayPal	Processing	0															
4	102	Customer_102	ORD-84355	12/23/2024	Tennis Racket	SPORTS	1	389.05	PayPal	Delivered	389.05															
5	103	Customer_103	ORD-57811	8/19/2025	Science	Books	5	233.92	PayPal	Processing	1169.6															
6	104	Customer_104	ORD-93614	10/26/2025	Biography	Books	1	552.51	Cash on Delivery	Processing	552.51															
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8	106	Customer_106	ORD-25885	2/2/2025	Blender	Home	0	978.63	Bank Transfer	Processing	0															
9	107	Customer_107	ORD-89122	1/3/2025	Biography	Books	5	587.68	PayPal	Returned	2938.4															
10	108	Customer_108	ORD-64400	10/23/2025	Science	Books	1	600.29	Cash on Delivery	Processing	600.29															
11	109	Customer_109	ORD-18512	5/3/2025	Tennis Racket	SPORTS	5	168.34	Credit Card	Shipped	841.7															
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13	111	Customer_111	ORD-72349	10/22/2025	Comics	Books	2	865.77	Credit Card	Returned	1731.54															
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18	116	Customer_116	ORD-63660	10/30/2025	Microwave	Home	4	0	Cash on Delivery	Processing	0															
19	117	Customer_117	ORD-72751	2/12/2025	Blender	ELECTRONICS	-2	10000	Cash on Delivery	Processing	-20000															
20	118	Customer_118	ORD-96061	5/31/2025	Science	Books	5	247.82	PayPal	Processing	1239.1															
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26	124	Customer_124	ORD-46136	5/31/2025	Comics	Books	5	0	PayPal	Cancelled	0															
27	125	Customer_125	ORD-23202	2/9/2025	Comics	Books	4	765.69	PayPal	Shipped	3062.76															
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32	130	Customer_130	ORD-34007	8/15/2025	Tennis Racket	SPORTS	2	0	Bank Transfer	Processing	0															
33	131	Customer_131	ORD-91396	6/1/2025	Lamp	Home	5	616.63	PayPal	Shipped	3083.15															
34	132	Customer_132	ORD-40186	4/7/2025	Yoga Mat	ELECTRONICS	1	266.7	PayPal	Delivered	266.7															
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38	136	Customer_136	ORD-20585	6/12/2025	Headphones	ELECTRONICS	1	696.71	Credit Card	Delivered	696.71															
39	137	Customer_137	ORD-91254	2/23/2025	Vacuum	Home	3	-100	Cash on Delivery	Delivered	-300															

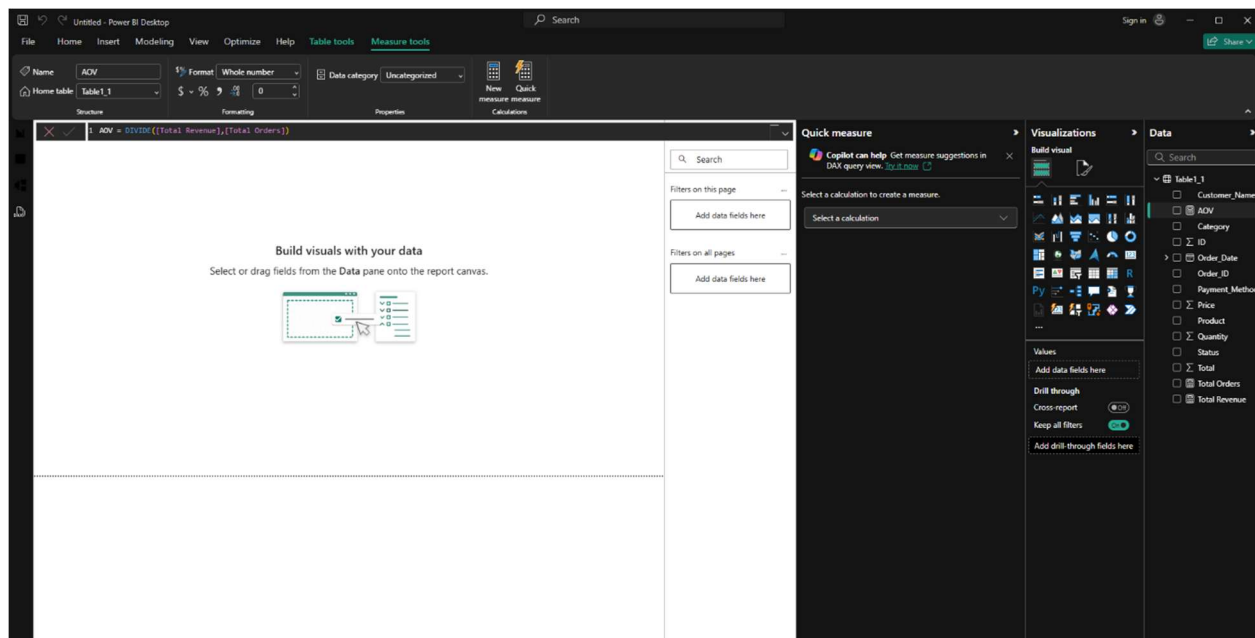
Data Transformation (Power Bi DAX)

After loading cleaned data into Power Bi:

The screenshot displays the Microsoft Power BI Desktop application. The main workspace shows a data model with a table named 'Table1'. The 'Table1' table has columns: Customer_Name, Order_ID, Order_Date, Order_Type, Product, Category, Quantity, Price, Payment_Method, Status, and Total. The 'Table1' table is currently selected, and its columns are visible in the 'Data' pane on the right. The 'Visualizations' pane on the left shows a 'Table' visual. The 'Formulas' pane on the right shows the DAX formula for the 'Order_Type' measure:

```
Order_Type =
IF (Table1[Total] < 0, "Return",
IF (Table1[Status] = "Cancelled", "Cancelled", "Sale"))
```

The 'Formulas' pane also shows the 'Table1' table structure with columns: Customer_Name, Order_ID, Order_Date, Order_Type, Product, Category, Quantity, Price, Payment_Method, Status, and Total. The 'Table' visual in the main workspace shows the data for the 'Table1' table, with columns: Customer_Name, Order_ID, Order_Date, Order_Type, Product, Category, Quantity, Price, Payment_Method, Status, and Total. The 'Table' visual is currently empty, showing only the column headers.



Measures Created in Power Bi:

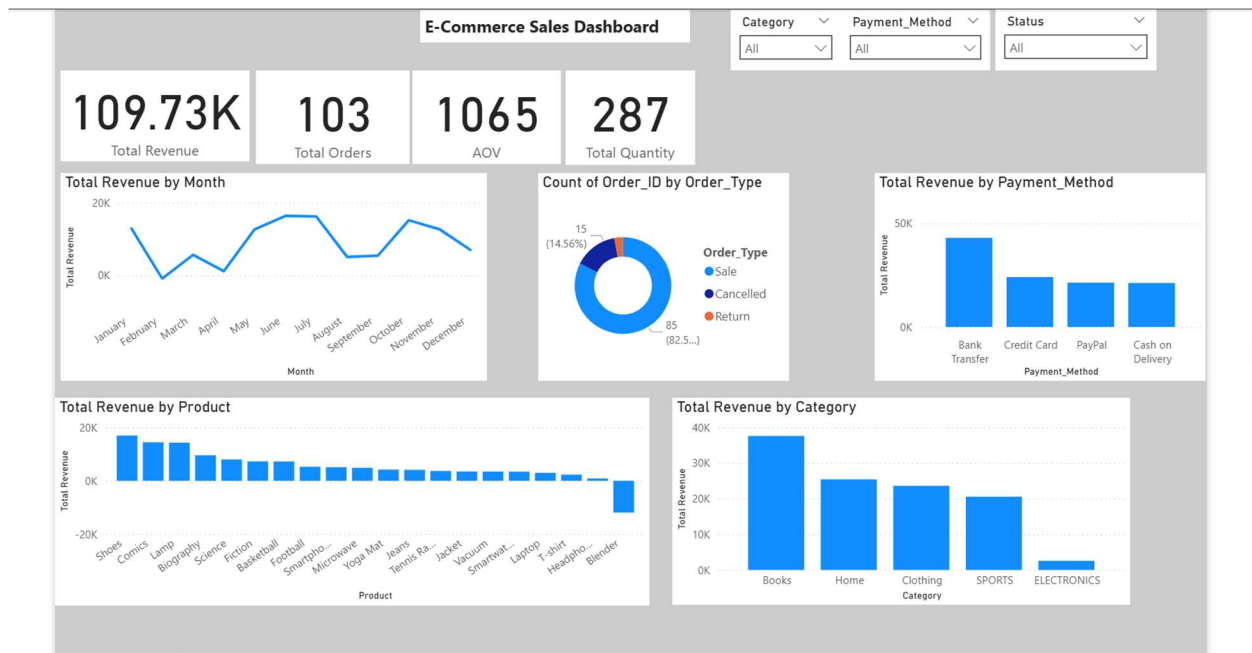
- Total Revenue = SUM(Total)
- Total Orders = COUNT(Order ID)
- Total Quantity = SUM(Quantity)
- Average Order Value (AOV) = Revenue / Orders

Calculated Column:

- Order_Type:
 - Return (negative values)
 - Cancelled
 - Sale

This helped in analyzing operational performance like returns and cancellations.

Dashboard Development (Power Bi)



An interactive Dashboard was built with:

KPI cards:

- Total Revenue
- Total Orders
- Average Order Value
- Total Quantity Sold

Visualizations:

- Sales Trend over time (Line chart)
- Product Performance (Bar chart)
- Category-wise Revenue (Bar chart)
- Order Status Distribution (Pie chart)
- Payment Method Analysis
- Customer-wise Revenue

Filters Slicers:

- Category
- Payment Method
- Status
- Order Date

These allow dynamic filtering and deeper exploration of data.

Key Findings:

1. Sales Trend Analysis

Revenue shows fluctuations across months, with noticeable peaks around mid-year and later months, indicating seasonal demand patterns

2. Order Type Distribution

Approx 82.5% of orders are successful sales, while the remaining consist of cancellations and returns, which still impact overall revenue.

3. Payment Method

Bank Transfer contributes the highest revenue, followed by Credit card and PayPal, indicating customer preference toward digital payment methods.

4. Category Performance

Books is the top-performing category, generating the highest revenue, while Electronics contributes the least.

5. Product Performance

Products like Shoes, Comics, and Lamps are among the top revenue generators, whereas some products (e.g. Blender) show negative revenue due to returns.

Business Insights:

- High revenue concentration in Books suggests an opportunity to expand offerings or marketing in this category.
- Negative revenue from certain products highlights the impact of returns, indicating potential quality or expectation issues.
- Strong reliance on Bank Transfer suggests optimizing and promoting this payment method further.
- Seasonal peaks indicate opportunities for targeted promotions and inventory planning.
- Reducing cancellations and returns can significantly improve overall profitability.

Final Conclusions:

The analysis highlights key revenue drivers, including product categories, customer purchasing patterns, and payment preferences. While the majority of transactions are

successful, returns and cancellations negatively affect overall performance. Leveraging high-performing categories and reducing operational inefficiencies can improve business outcomes.